

Challenge (Habanero)

- Q1.** Factor the trinomial $x^2 + 5x + 6$ to optimize a gaming scoreboard's algorithm.
(A) $x^2 + 2x + 3x + 6$
(B) $(x + 2)(x + 3)$
(C) $(x + 1)(x + 4)$
(D) $(x - 2)(x - 3)$
(E) $(x - 1)(x - 4)$
- Q2.** A coder needs to factor $2x^2 + 7x + 3$ for a data usage tracker. What is the factored form?
(A) $(2x + 1)(x + 3)$
(B) $(x + 1)(2x + 3)$
(C) $(x - 1)(2x - 3)$
(D) $(2x - 1)(x - 3)$
(E) $(2x + 3)(x + 1)$
- Q3.** What is the factored form of $x^2 - 4x - 5$ for a video game's scoring system?
(A) $(x + 1)(x - 5)$
(B) $(x - 1)(x + 5)$
(C) $(x - 5)(x + 1)$
(D) $(x + 5)(x - 1)$
(E) $(x - 4)(x + 1)$
- Q4.** A tech company uses $3x^2 + 10x + 8$ in their data analysis. What is the factored form?
(A) $(3x + 2)(x + 4)$
(B) $(3x + 4)(x + 2)$
(C) $(x + 2)(3x + 4)$
(D) $(3x - 2)(x - 4)$
- (E) $(x - 2)(3x - 4)$
- Q5.** To optimize a gaming algorithm, factor $x^2 - 6x + 8$.
(A) $(x + 2)(x - 4)$
(B) $(x + 4)(x - 2)$
(C) $(x - 2)(x - 4)$
(D) $(x - 4)(x + 2)$
(E) $(x + 1)(x - 8)$
- Q6.** A coder needs to factor $x^2 + 2x - 6$ for a scoreboard's logic. What is the factored form?
(A) $(x - 2)(x + 3)$
(B) $(x + 2)(x - 3)$
(C) $(x + 6)(x - 1)$
(D) $(x + 1)(x + 6)$
(E) $(x - 1)(x - 6)$
- Q7.** A data analyst uses $2x^2 - 5x - 3$ in their coding. What is the factored form?
(A) $(2x + 1)(x - 3)$
(B) $(x + 1)(2x - 3)$
(C) $(x - 1)(2x + 3)$
(D) $(2x - 1)(x + 3)$
(E) $(2x + 3)(x - 1)$
- Q8.** To debug a gaming code, factor $x^2 - 3x - 10$.
(A) $(x + 5)(x - 2)$
(B) $(x + 2)(x - 5)$
(C) $(x + 1)(x - 10)$
(D) $(x - 1)(x + 10)$
(E) $(x + 10)(x - 1)$

Open Ended Questions (FRQ)

- Q9.** Factor the trinomial $4x^2 + 11x + 6$ and explain how it can be applied in a coding logic for a gaming scoreboard.
- Q10.** Factor $3x^2 - 7x - 2$ and describe a scenario in data analysis where this factored form can be utilized.

Chef's Tips

- Remind students to check their signs when factoring trinomials.
- Encourage students to use different methods for factoring, such as grouping or using the quadratic formula.
- Emphasize the importance of factoring in real-world applications, such as data usage tracking and coding logic.